Intermediate Elective 2

Minh Tri Ngo

Set Up

Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrapping axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# visualizing missing data
library(naniar)

# arranging plots
library(patchwork)

# imported libraries from example
library(showtext)
library(camcorder)
library(ggtext)
library(glue)
library(ggfx)
```

Data

```r
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                    sheet = "Aquatic Insects")

# site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                   sheet = "sites")

# water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling
↪  Data-2026-03-10.xlsx"),
                           sheet = "Water Quality",
                           na = c("", "n/a", "N/A", "over", "173+"))


# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS sites sampled four times a year
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")



# creating new clean object from taxon_list
taxon_list_clean <- taxon_list |>
  # converting taxon name to snake case (no spaces or capital letters,
  # only underscores)
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))

# creating new clean object from water quality
water_quality_clean <- water_quality |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites in ncos_sites
  semi_join(ncos_sites, by = "site") |>
  # create new column to join with later data frames
  unite("date_site", date, site, remove = TRUE) |>
  # group by date_site column
```

```r
  group_by(date_site) |>
  # calculate median pH, salinity, DO
  summarize(med_ph = median(p_h, na.rm = TRUE),
            med_sal = median(salinity_ppt, na.rm = TRUE),
            med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  # ungroup data frame
  ungroup() |>
  # filter out salinity outliers
  filter(date_site != "2022-08-12_NVBR")


# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites occurring in ncos_sites
  semi_join(ncos_sites, by = c("site")) |>
  # renaming columns
  rename(date = date_on_vial) |>
  # select columns of interest
  select(date, sample_type, site,
         ostracod,
         copepod,
         hemiptera_corixidae_boatman,
         cladocera,
         nematode,
         diptera_ceratopogonidae,
         annelida_oligochaete,
         diptera_chironomid,
         annelida_polychaete) |>
  # filter to only include "filtered beaker" samples
  filter(sample_type %in% c("FB 250", "FB250")) |>
  # convert the data frame to long format
  pivot_longer(cols = ostracod:annelida_polychaete,
               names_to = "taxon",
               values_to = "abundance") |>
  # group by date, site, taxon
  group_by(date, site, taxon) |>
  # calculate average abundance per liter (sum abundance divided by 7.5
  ↪  liters)
  summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
  # ungroup data frame
  ungroup() |>
```

```r
  # create a new column called `date_site` to join with water quality
  unite("date_site", date, site, remove = FALSE) |>
  # join with water quality
  left_join(water_quality_clean, by = c("date_site")) |>
  # join with taxon list
  left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
  # add full names of sites
  mutate(site_full = case_when(
    site == "NVBR" ~ "North of Venoco Bridge",
    site == "NEC" ~ "South of Venoco Bridge",
    site == "NMC" ~ "Main Channel",
    site == "NPB" ~ "South of Phelps Bridge",
    site == "NPB1" ~ "North of Phelps Bridge",
    site == "NPB2" ~ "Phelps Road",
    site == "NWP" ~  "West Pond",
    site == "NDC" ~ "Devereux Creek"
  )) |>
  # setting factor levels for site
  mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm
   ↪ = TRUE),
         site_full = fct_rev(site_full))

# creating new object from clean aquatic inverts
copepods_df <- aq_ins_clean |>
  # filter to only include copepods
  filter(taxon == "copepod") |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))

# creating new clean object from aquatic inverts
ab_do <- aq_ins_clean |>
  # log transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit+1),
         # convert taxon to title
         taxon = to_any_case(taxon, case = "title"),
         # order taxon levels by abudance
         taxon = fct_reorder(taxon, ave_lit, .fun="max"))
```

1. Explain Inspiration

- The visualization I chose for my inspiration was the "Can an exploding snowman predict the summer season" by Nicola Rennie.
- I felt that this visualization makes sense as I would like to analyze temperature in relation to the different taxon availability and the graph allows me to analyze and communicate my findings the msot efficiently.
- The variables that I will choose in my dataset to show my visualization will be dissolved, taxon, and average abundance. In terms of visual components, the dissolved will be on the x axis. Taxon will have different colors to differentiate. And the y axis will have the average abundance.

2. Plan figure

3. Code figure

```
# Load fonts ----------------------------------------------------------------

font_add_google("Poppins", "Poppins")
showtext_auto()
showtext_opts(dpi = 300)
title_font <- "Poppins"
body_font <- "Poppins"


# Define colours and fonts--------------------------------------------------

bg_col <- "#F2F4F8"
text_col <- "#151C28"
highlight_col <- "cornflowerblue"

# Taxon color palette (one color per taxon)
taxon_colors <- c(
  "Annelida Oligochaete"          = "#2D6A4F",
  "Annelida Polychaete"           = "#52B788",
  "Cladocera"                     = "#74C69D",
  "Copepod"                       = "#B7E4C7",
  "Diptera Ceratopogonidae"       = "#D4A373",
  "Diptera Chironomid"            = "#A0522D",
```

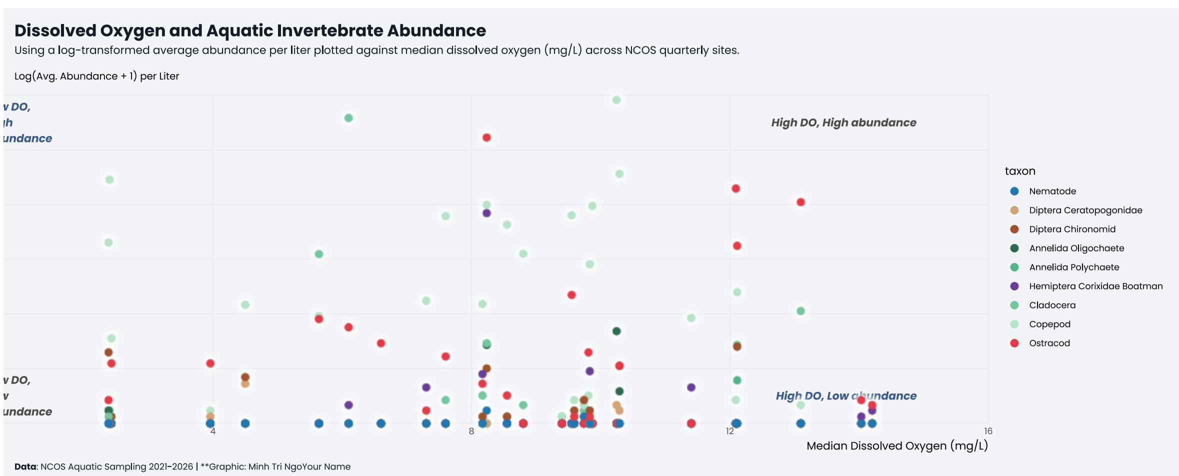
```r
  "Hemiptera Corixidae Boatman"   = "#6A3D9A",
  "Nematode"                      = "#1D78B4",
  "Ostracod"                      = "#E63946"
)


# Data prep ------------------------------------------------------------

plot_data <- ab_do |>
  filter(!is.na(med_do), !is.na(log_ave_lit))

med_do_val  <- median(plot_data$med_do,      na.rm = TRUE)
med_ab_val  <- median(plot_data$log_ave_lit, na.rm = TRUE)


# Start recording ------------------------------------------------------

gg_record(
  dir = file.path("2025", "2025-12-02", "recording"),
  device = "png",
  width = 7,
  height = 6,
  units = "in",
  dpi = 300
)


# Define text ----------------------------------------------------------

social <- "Your Name"
icon <- "<span
↪  style='font-size:8pt;font-family:\"FA\";color:#3C598E;'>&#xf111;</span> "
title <- "Dissolved Oxygen and Aquatic Invertebrate Abundance"
st <- glue("Using a log-transformed average abundance per liter plotted
↪  against median dissolved oxygen (mg/L) across NCOS quarterly sites.")
cap <- paste0(
  "**Data**: NCOS Aquatic Sampling 2021-2026 | **Graphic: Minh Tri Ngo",
   ↪  social
)

most_abundant <- ab_do |>
  slice_max(log_ave_lit)
```

```r
highest_do <- ab_do |>
  slice_max(med_do)

# Plot --------------------------------------------------------------------

ggplot(
  data = ab_do        # <-- swapped: was `plot_data`
) +
  annotate(
    "rect",
    xmin = median(ab_do$med_do), xmax = Inf,        # <-- swapped: was
     ↪  `plot_data$duration`
    ymin = median(ab_do$log_ave_lit), ymax = Inf, alpha = 0.5,   # <--
     ↪  swapped: was `plot_data$tre200m0`
    fill = "grey"
  ) +
  annotate(
    "text",
    x = 15, y = 5.5, label = "High DO, High abundance",          # <--
     ↪  swapped: coordinates + label
    colour = "grey30", family = body_font, fontface = "bold.italic",
    hjust = 1.05
  ) +
  annotate(
    "rect",
    xmin = median(ab_do$med_do), xmax = Inf,        # <-- swapped
    ymin = -Inf, ymax = median(ab_do$log_ave_lit), alpha = 0.5,  # <--
     ↪  swapped
    fill = highlight_col
  ) +
  annotate(
    "text",
    x = 15, y = 0.5, label = "High DO, Low abundance",           # <--
     ↪  swapped
    colour = "#3C598E", family = body_font, fontface = "bold.italic",
    hjust = 1.05
  ) +
  annotate(
    "rect",
    xmin = -Inf, xmax = median(ab_do$med_do),       # <-- swapped
    ymin = median(ab_do$log_ave_lit), ymax = Inf, alpha = 0.5,   # <--
     ↪  swapped
```

```
  fill = highlight_col
) +
annotate(
  "text",
  x = 0.5, y = 5.5, label = "Low DO,\nHigh\nabundance",        # <--
   ↪ swapped
  colour = "#3C598E", family = body_font, fontface = "bold.italic",
  hjust = 0
) +
annotate(
  "rect",
  xmin = -Inf, xmax = median(ab_do$med_do),       # <-- swapped
  ymin = -Inf, ymax = median(ab_do$log_ave_lit), alpha = 0.5,  # <--
   ↪ swapped
  fill = "grey"
) +
annotate(
  "text",
  x = 0.5, y = 0.5, label = "Low DO,\nLow\nabundance",         # <--
   ↪ swapped
  colour = "grey30", family = body_font, fontface = "bold.italic",
  hjust = 0
) +
with_outer_glow(
  geom_point(
    mapping = aes(
      x = med_do, y = log_ave_lit,    # <-- swapped: was `duration`,
       ↪ `tre200m0`
      colour = taxon                  # <-- swapped: was `correct`; removed
       ↪ `shape`
    ),
    size = 3
  ),
  colour = "white",
  expand = 3
) +
scale_colour_manual(values = c(      # <-- swapped: your 9 taxon colors
  "Annelida Oligochaete"        = "#2D6A4F",
  "Annelida Polychaete"         = "#52B788",
  "Cladocera"                   = "#74C69D",
  "Copepod"                     = "#B7E4C7",
  "Diptera Ceratopogonidae"     = "#D4A373",
```

```r
  "Diptera Chironomid"           = "#A0522D",
  "Hemiptera Corixidae Boatman" = "#6A3D9A",
  "Nematode"                     = "#1D78B4",
  "Ostracod"                     = "#E63946"
)) +
scale_x_continuous(limits = c(0, 16)) +    # <-- swapped: DO range ~0-15
↪  mg/L
scale_y_continuous(limits = c(0, 6)) +     # <-- swapped: log(ave_lit+1)
↪  range
labs(
  x = "Median Dissolved Oxygen (mg/L)",    # <-- swapped
  y = "Log(Avg. Abundance + 1) per Liter", # <-- swapped
  title = title, subtitle = st,
  caption = cap
) +
coord_cartesian(expand = FALSE, clip = "off") +
theme_minimal(base_size = 10, base_family = body_font) +
theme(
  legend.position = "right",               # <-- swapped: "none" → "right"
   ↪  to show taxon legend
  plot.margin = margin(10, 20, 10, 10),
  plot.title.position = "plot",
  plot.caption.position = "plot",
  plot.background = element_rect(fill = bg_col, colour = bg_col),
  panel.background = element_rect(fill = bg_col, colour = bg_col),
  plot.title = element_textbox_simple(
    colour = text_col,
    hjust = 0,
    halign = 0,
    margin = margin(b = 0, t = 5),
    family = title_font,
    face = "bold",
    size = rel(1.5)
  ),
  plot.subtitle = element_textbox_simple(
    colour = text_col,
    hjust = 0,
    halign = 0,
    margin = margin(b = 35, t = 5),
    family = body_font
  ),
  plot.caption = element_textbox_simple(
    colour = text_col,
```

```
      hjust = 0,
      halign = 0,
      margin = margin(b = 0, t = 10),
      family = body_font
    ),
    strip.text = element_textbox_simple(
      face = "bold",
      margin = margin(t = 10),
      size = rel(0.9)
    ),
    panel.grid.minor = element_blank(),
    axis.title.x = element_text(hjust = 1, margin = margin(t = 5)),
    axis.title.y.left = element_text(
      angle = 0, vjust = 1.07,
      family = body_font,
      size = rel(0.9),
      margin = margin(r = -215)
    )
  )
```




```
ggplot(
  data = plot_data
) +
  annotate(
    "rect",
    xmin = med_do_val, xmax = Inf,
    ymin = med_ab_val, ymax = Inf, alpha = 0.5,
    fill = "grey"
```

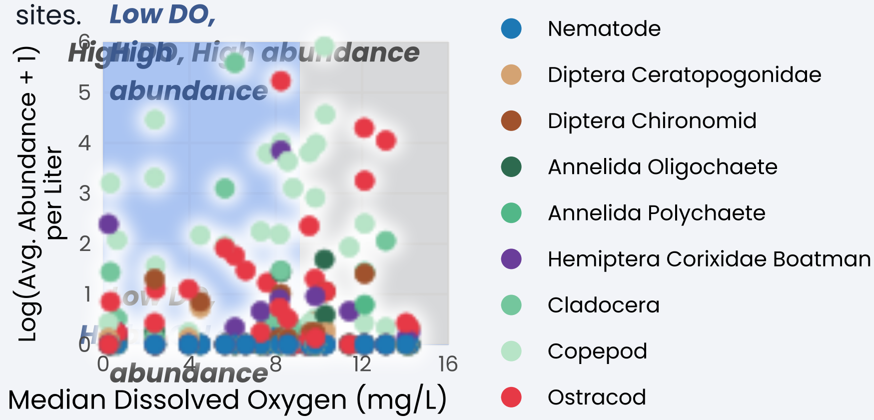
```
) +
annotate(
  "text",
  x = 15.5, y = 5.8, label = "High DO, High abundance",
  colour = "grey30", family = body_font, fontface = "bold.italic",
  hjust = 1.05
) +
annotate(
  "rect",
  xmin = med_do_val, xmax = Inf,
  ymin = -Inf, ymax = med_ab_val, alpha = 0.5,
  fill = highlight_col
) +
annotate(
  "text",
  x = 15.5, y = 0.2, label = "High DO, Low abundance",
  colour = "#3C598E", family = body_font, fontface = "bold.italic",
  hjust = 1.05
) +
annotate(
  "rect",
  xmin = -Inf, xmax = med_do_val,
  ymin = med_ab_val, ymax = Inf, alpha = 0.5,
  fill = highlight_col
) +
annotate(
  "text",
  x = 0.3, y = 5.8, label = "Low DO,\nHigh\nabundance",
  colour = "#3C598E", family = body_font, fontface = "bold.italic",
  hjust = 0
) +
annotate(
  "rect",
  xmin = -Inf, xmax = med_do_val,
  ymin = -Inf, ymax = med_ab_val, alpha = 0.5,
  fill = "grey"
) +
annotate(
  "text",
  x = 0.3, y = 0.2, label = "Low DO,\nLow\nabundance",
  colour = "grey30", family = body_font, fontface = "bold.italic",
  hjust = 0
) +
```

```r
with_outer_glow(
  geom_point(
    mapping = aes(
      x = med_do, y = log_ave_lit,
      colour = taxon
    ),
    size = 3
  ),
  colour = "white",
  expand = 3
) +
scale_colour_manual(values = c(
  "Annelida Oligochaete"        = "#2D6A4F",
  "Annelida Polychaete"         = "#52B788",
  "Cladocera"                   = "#74C69D",
  "Copepod"                     = "#B7E4C7",
  "Diptera Ceratopogonidae"     = "#D4A373",
  "Diptera Chironomid"          = "#A0522D",
  "Hemiptera Corixidae Boatman" = "#6A3D9A",
  "Nematode"                    = "#1D78B4",
  "Ostracod"                    = "#E63946"
)) +
scale_x_continuous(limits = c(0, 16)) +
scale_y_continuous(limits = c(0, 6)) +
labs(
  x = "Median Dissolved Oxygen (mg/L)",
  y = "Log(Avg. Abundance + 1)\nper Liter",
  title = title, subtitle = st,
  caption = cap
) +
coord_cartesian(expand = FALSE, clip = "off") +
theme_minimal(base_size = 10, base_family = body_font) +
theme(
  legend.position = "right",
  plot.margin = margin(10, 20, 10, 50),  # <-- increased left margin to
   ↪  make room for y title
  plot.title.position = "plot",
  plot.caption.position = "plot",
  plot.background = element_rect(fill = bg_col, colour = bg_col),
  panel.background = element_rect(fill = bg_col, colour = bg_col),
  plot.title = element_textbox_simple(
    colour = text_col,
    hjust = 0,
```

```
    halign = 0,
    margin = margin(b = 0, t = 5),
    family = title_font,
    face = "bold",
    size = rel(1.5)
  ),
  plot.subtitle = element_textbox_simple(
    colour = text_col,
    hjust = 0,
    halign = 0,
    margin = margin(b = 10, t = 5),  # <-- reduced from 35 to 10
    family = body_font
  ),
  plot.caption = element_textbox_simple(
    colour = text_col,
    hjust = 0,
    halign = 0,
    margin = margin(b = 0, t = 10),
    family = body_font
  ),
  strip.text = element_textbox_simple(
    face = "bold",
    margin = margin(t = 10),
    size = rel(0.9)
  ),
  panel.grid.minor = element_blank(),
  axis.title.x = element_text(hjust = 1, margin = margin(t = 5)),
  axis.title.y.left = element_text(
    angle = 90,         # <-- changed from 0 to 90 (normal upright
     ↪  rotation)
    vjust = 0.5,        # <-- changed from 1.07
    family = body_font,
    size = rel(0.9),
    margin = margin(r = 5)  # <-- changed from -215 to 5
  )
)
```

Dissolved Oxygen and Aquatic Invertebrate Abundance

Using a log-transformed average abundance per liter plotted against median dissolved oxygen (mg/L) across NCOS quarterly sites.



Data: NCOS Aquatic Sampling 2021-2026 | **Graphic: Minh Tri NgoYour Name